Topic 11 — Population (जनसंख्या) Geography

Covers syllabus (click to expand)

Census of India | Factors of Distribution | Population Density (arithmetic / physiological / agricultural) | Growth Vocabulary | Sex Ratio | Child Sex Ratio | Literacy | Birth–Death–Natural Growth–TFR | Migration (streams / push–pull) | Composition (age / occupation / SC–ST) | Urbanization | Million Cities | Population Pyramid (पिरामिड) | Demographic Transition & Theories

Sources baked in: Census 2011 (official until Census 2027), NCERT Class 12 (Population), PWOnlyIAS / Drishti notes, NFHS-4/5, NPP 2000, UPPCS Prelims 2018–2025 **Weight:** ★★★★ — census firsts, 2011 number facts, UP extremes, TFR vs CBR trap, DTT/theories **Last verified:** August 2026 **Current Affairs:** UNFPA World Population Report; India most populous in UN estimates ~2023 — **Census 2011 figures stay frozen** until Census 2027; NFHS-5 TFR ~2.0 (NFHS-4 / 2018 figure remains **2.2**)

Current Affairs (this topic)

Year	Fact	Why asked	Source
Census 2011	Official number set until **Census 2027**	Every density/SR/literacy MCQ	ORGI
NFHS-5	TFR ~**2.0** nationally	NFHS-4 (2015–16) figure used in 2018 was **2.2**	MoHFW
UNFPA	World Population Report	2021 Q79	UNFPA
~2023	UN estimate: India most populous	CA only — 2011 still 2nd in Census table	UN / UNFPA
NPP 2000	Stable population **2045**	Policy year fact	MoHFW
11 July	World Population Day	2018 Q114	UN

N.0 Framework (UPSC (संघ लोक सेवा आयोग) / NCERT map first)

Population geography asks **how many people live where**, **how that number changes**, and **what structure** the population has.

- **Distribution** is the spread of people across space.
- **Density** is persons per unit area (with special NCERT variants for agricultural pressure).
- **Composition** covers age, sex, literacy, rural–urban status, religion, caste/tribe and occupation.
- **Dynamics** covers births, deaths, natural growth, fertility and migration.

- India's official Prelims number set remains **Census 2011** until **Census 2027** results replace it.
- UN estimates may say India became the world's most populous country around **2023**, but that does **not** rewrite Census 2011 tables in the hall.

Core 2011 Indias

- Population ≈ **121.09 crore**.
- Density **382**, sex ratio **943**, child sex ratio **919**, literacy **74.04%** (age **7+**), urban share **31.16%**.
- Decadal growth for **2001–11** was **17.64%**.
- Composition note: Scheduled Castes about **16.6%**; Scheduled Tribes about **8.6%**.

N.1 Census of India

A census is the official count of population and its attributes, taken at fixed intervals.

- India's first census effort is dated **1872** as a **non-synchronous** count; the **2023** UPPCS key used option **1871**.
- The first **synchronous** all-India census was **1881**, and the series then ran every **ten years**.
- **1921** is the **Great Divide** — the first decade of decline in the growth rate after the modern census series.
- **Census 2011** is the **15th** census and the **7th** after Independence.
- India's 2011 population was **1,210,854,977** ≈ **121.09 crore**, about **one-sixth** of the world.
- Decadal growth **2001–11** was **17.64%**, down from **21.54%** in **1991–2001**.
- **Nagaland** recorded **negative** state-level growth in 2001–11.
- **Sikkim** is the **least populous** state.

NCERT growth phases (1901–2011)

Phase	Years	Character
I	1901–1921	Stagnant / low growth (ends at the Great Divide)
II	1921–1951	Steady growth
III	1951–1981	Rapid growth / population explosion
IV	1981–2011	High growth with a **declining rate**

> **Logic:** ** Teach **1872** historically and **1881** as first sync. The UPPCS **2023** answer option was **1871**.

Inline PYQ — UPPCS Prelims 2023, Q128

When was the Census conducted for the first time in India?

A. 1850

B. 1861

C. 1871

D. 1881

Show answer

Ans: C — The non-synchronous census year keyed here is **1871**. The historical first non-synchronous census is usually dated **1872**; the first synchronous all-India census is **1881**.

N.2 Population Distribution

People are unevenly spread across India. About half of India's population is concentrated in a few large states led by **Uttar Pradesh, Maharashtra, Bihar, West Bengal and Andhra Pradesh**. The 2011 Andhra figure is for the undivided state.

Factors affecting distribution

Physical factors:

- **Terrain** — plains favour dense settlement; steep Himalaya and many NE hills remain sparse.
- **Climate** — moderate (नरम दल) monsoon belts attract people; extreme cold, arid or very wet zones deter dense living.
- **Soil** — fertile alluvium of the Ganga plain and many deltas supports intensive farming and denser villages.
- **Water** — perennial rivers and irrigation raise carrying capacity.

Socio-economic and historical factors:

- **Settled agriculture and early civilisation** kept river plains crowded for centuries.
- **Minerals and industry** pull people toward mining–industrial belts (ple parts of Chotanagpur (छोटानागपुर) and coastal industrial hubs).
- **Transport networks** thicken settlement along roads and railways.
- **Urbanisation and jobs** concentrate migrants in metros such as Delhi, Mumbai, Kolkata, Bengaluru, Chennai, Hyderabad (हैदराबाद) and Ahmedabad (अहमदाबाद).

Map pair

- Crowded belts: Indo-Gangetic plain and many coastal plains.
- Sparse belts: Himalaya, north-eastern hills, Thar, and some dry or rugged interior pockets.
- Census 2011 descending population among the states papers reuse most often is **Uttar Pradesh > Maharashtra > Bihar > West Bengal > Andhra Pradesh**.
- A 2024 question listing Bihar, Andhra Pradesh, Uttar Pradesh and West Bengal is answered as **UP > Bihar > WB > AP**.

Inline PYQ — UPPCS Prelims 2024, Q5

Write in descending order the following States on the basis of their population as per Census, 2011:

1. Bihar

2. Andhra Pradesh
3. Uttar Pradesh
4. West Bengal

A. 3, 4, 1, 2

B. 1, 3, 2, 4

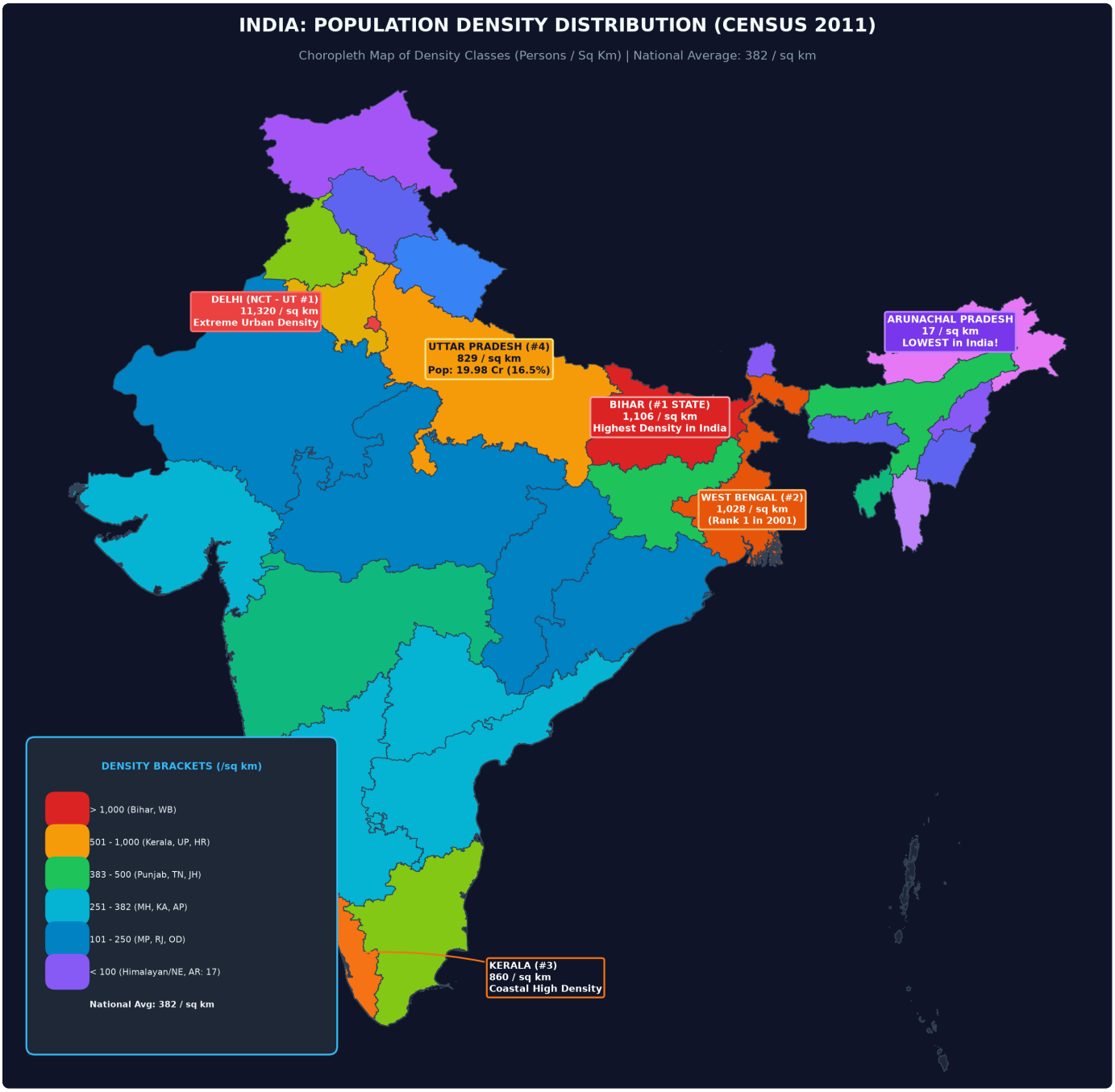
C. 1, 3, 4, 2

D. 3, 1, 4, 2

Show answer

Ans: D — UP > Bihar > WB > AP (2011 AP undivided).

N.3 Population Density



Population density compares people with land. Prelims usually quotes **arithmetic density**, but NCERT also teaches two agricultural-pressure measures.

Type	Formula	What it shows
Arithmetic density	Total population ÷ total area	Everyday “persons per km²” fact
Physiological density	Total population ÷ **net sown** area	Pressure of people on cultivated land
Agricultural density	Agricultural population ÷ **net sown** area	Pressure of farm workers on cultivated land

- India’s Census **2011** arithmetic density is **382** (do not mix with **2001 ≈ 325**).
- High-density **state** ladder note: **Bihar (~1106)** > **West Bengal (~1028)** > **Kerala (~860)** > **Uttar Pradesh (829)**. Uttar Pradesh **829** sits well above India’s **382**.

- The lowest density **state** is **Arunachal Pradesh (~17)**.
- NCT **Delhi** is a very high-density UT, but it is **not** a “state” answer for lowest density.
- Moderate densities appear in many peninsular and central states; Himalayan and several NE hill states stay relatively low.

Inline PYQ — UPPCS Prelims 2021, Q142

Which of the following State has the lowest density of population as per the Census of 2011?

A. Himachal (हिमाचल) Pradesh

B. Arunachal Pradesh

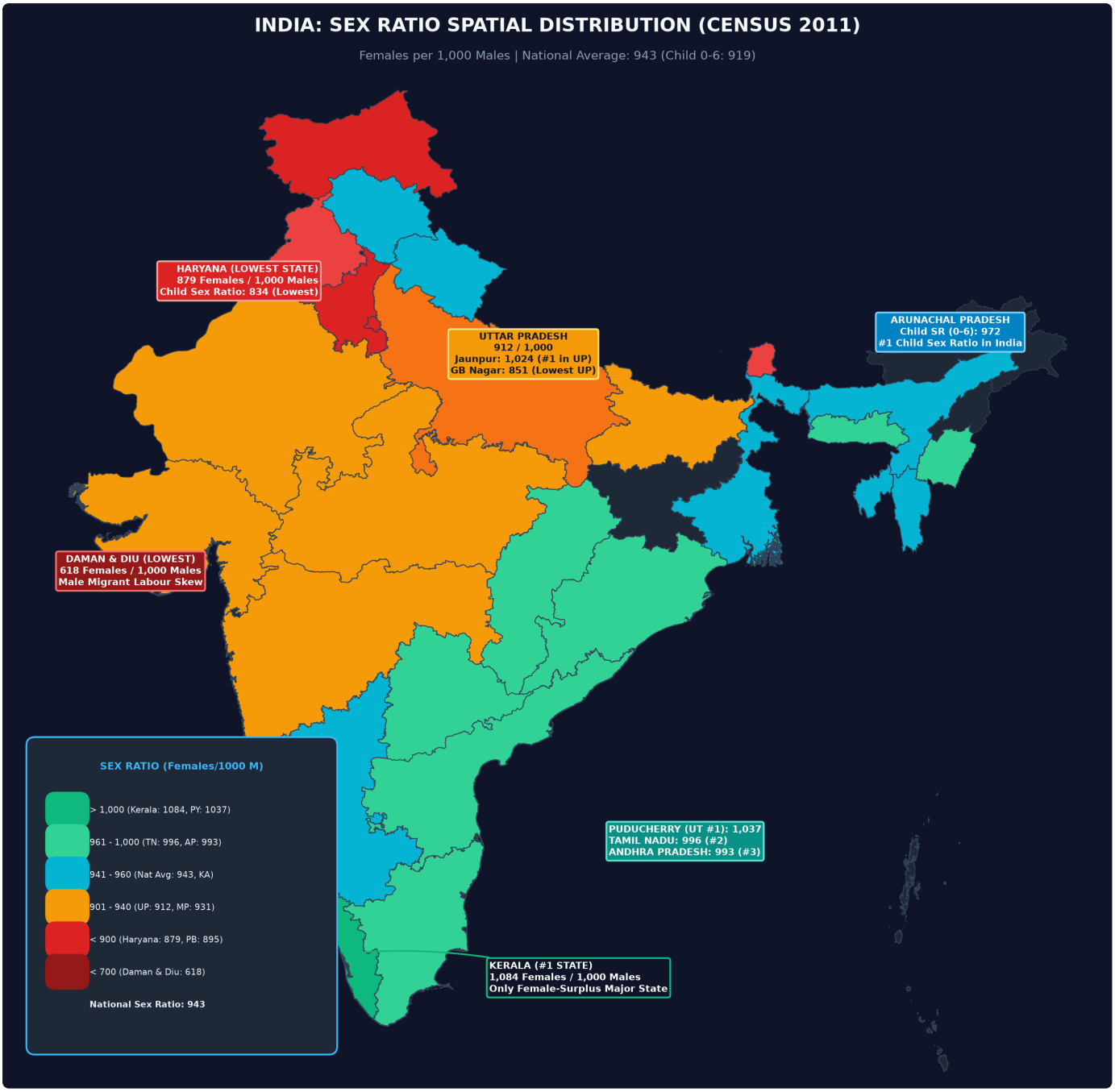
C. West Bengal

D. Mizoram

Show answer

Ans: B — Arunachal Pradesh is the lowest-density **state**.

N.4 Sex Ratio



Sex ratio is the number of females per **1,000** males in the total population.

- Formula used in Census notes: **(females ÷ males) × 1,000**.
- India's Census **2011** sex ratio is **943** (up from **933** in 2001).
- The highest State sex ratio in 2011 is **Kerala** (**1,084**).
- Among major States, **Tamil Nadu (नाडु)** and **Andhra Pradesh** also sit high on the 2011 list.
- Among States in the usual 2011 set, the lowest sex ratio is **Haryana** (**879**).
- Some Union Territories show even lower figures than Haryana, but State-level stems usually expect **Haryana**.
- India's **rural** sex ratio is higher than the **urban** sex ratio in the 2011 pattern, because male-selective migration into cities skews urban counts.
- Uttar Pradesh sex ratio in 2011 is **912**, below India's **943**.

- Within UP, district notes often cite **Jaunpur** among the better sex-ratio districts.

Inline PYQ — UPPCS Prelims 2018, Q38

Which of the following States is ranked economically at the top but the lowest on the basis of sex-ratio?

A. Madhya Pradesh

B. Maharashtra

C. West Bengal

D. Haryana

Show answer

Ans: D — Haryana.

N.5 Child Sex Ratio

Child sex ratio (CSR) counts girls per 1,000 boys in the age group **0–6**.

- Formula: **(girls aged 0–6 ÷ boys aged 0–6) × 1,000**.
- India's Census **2011** CSR is **919**, worse than **927** in 2001.
- CSR is a sharper alarm than overall sex ratio because it reflects recent births and early childhood survival, not lifetime migration alone.
- The State with the **lowest CSR in both rural and urban areas** in the usual 2011 set is **Haryana**.
- Do not mark Uttar Pradesh, Kerala or Jammu & Kashmir for that lowest-CSR question.
- Southern and several north-eastern States generally show healthier CSR patterns than the north-western belt in Census maps.
- Uttar Pradesh's overall sex ratio (**912**) is already below India's **943**. India's CSR is **919**. CSR stems still treat **Haryana** as the classic lowest rural-plus-urban answer.

Inline PYQ — UPPCS Prelims 2018, Q110

According to 2011 census, which of the following states has the lowest child sex-ratio both in Rural and Urban areas?

A. Uttar Pradesh

B. Kerala

C. Haryana

D. Jammu & Kashmir

Show answer

Ans: C — **Haryana**.

N.6 Literacy

Census literacy covers persons aged **7 years and above** who can read and write with understanding in any language.

- That age cut-off defines the **effective literacy rate**. Counting the whole population including children below 7 gives a lower **crude literacy rate**.
- India's Census **2011** literacy is **74.04%** (male **82.14%**, female **65.46%**).
- The gender gap is therefore about **16.7** percentage points in the 2011 set.
- Highest literacy State (सर्वाधिक साक्षरता) is **Kerala** (about **94%**).
- Lowest literacy State among major States is **Bihar** (about **61.8%**).
- Literacy rose from about **64.8%** in 2001 to **74.04%** in 2011, with a large absolute gain among rural females.
- Urban literacy is higher than rural literacy in every major State pattern taught from Census 2011.
- In Uttar Pradesh, overall literacy in 2011 is about **67.7%**, below India's **74.04%**.
- In Uttar Pradesh, the lowest **female literacy** district fact is **Shrawasti**.
- Do not mark Rampur, Balrampur or Bahraich for that UP female-literacy floor question.
- The **Lorenz curve** measures **income inequality**, not illiteracy.

Inline PYQ — UPPCS Prelims 2018, Q105

Which of the following districts of Uttar Pradesh has the lowest female literacy rate as per 2011 census?

A. Rampur

B. Balrampur

C. Shrawasti

D. Bahraich

Show answer

Ans: C — **Shrawasti**.

N.7 Birth Rate, Death Rate, Natural Growth, TFR

Growth vocabulary

- **Absolute growth** is the difference between population at two dates ($P_2 - P_1$).
- **Growth rate** expresses that change as a **percentage** of the base population.
- **Natural growth** comes from births and deaths only: **CBR − CDR**.
- **Induced growth** comes from **migration** (people moving in or out).
- Total population change is natural growth plus the migration balance.

Rates and fertility

- **Crude Birth Rate (CBR)** and **Crude Death Rate (CDR)** are live births and deaths per **1,000** mid-year population.
- Migration and marriages are **not** part of natural growth.
- **Total Fertility Rate (TFR)** is the average number of children **per woman**.
- Replacement-level TFR is **2.1**.
- NFHS-4 (2015–16) TFR was **2.2** (the figure used in the 2018 paper).
- NFHS-5 national TFR is about **2.0**. That is a later survey figure; it does not change the 2018 paper answer.
- **National Population Policy 2000** aimed for replacement fertility and a **stable population by 2045**.
- **World Population Day** is **11 July**.
- The **World Population Report** is released by **UNFPA**, not IMF, WHO or UNDP (यूएनडीपी).

UP Population Control Bill 2021 trap

- The 2021 paper used wording like “2.1 / 2.7 **per thousand**” for fertility.
- That unit belongs to **CBR**, not **TFR**, so both statements fail.

Inline PYQ — UPPCS Prelims 2018, Q34

Natural growth of population is the outcome of which of the following?

1. Crude Birth Rate
2. Crude Death Rate
3. Migration
4. Marriages

Select the correct answer from the code given below:

A. Only 1

B. Only 3

C. 2 and 4

D. 1 and 2

Show answer

Ans: D — Natural growth = **CBR – CDR**. Migration is not “natural” growth.

Inline PYQ — UPPCS Prelims 2020, Q117

Replacement level Total Fertility Rate (TFR) is:

A. 1.5

B. 1.8

C. 2.1

D. 2.5

Show answer

Ans: C — Replacement TFR = **2.1** children per woman.

Inline PYQ — UPPCS Prelims 2021, Q4

With reference to the Uttar Pradesh Population Control Bill 2021, which of the following statements is/are correct?

1. This Bill aims to bring the birth rate down to 2.1 per thousands population by 2026.
2. At present, the fertility rate in the State is 2.7 per thousand.

A. Only 1

B. Only 2

C. Both 1 and 2

D. Neither 1 nor 2

Show answer

Ans: D — TFR / replacement **2.1** is not a CBR “per thousand” figure.

N.8 Migration

Migration in the Census means a change of residence. It changes both **size** and **composition** of population at origin and destination.

- The four (चातुर्याम) internal streams are rural-to-rural, rural-to-urban, urban-to-urban and urban-to-rural.
- India’s **largest** internal migration stream is **rural-to-rural**.
- Female migration is classically linked with **marriage**.
- Male migration is classically linked with **work / employment**.
- **Immigration** means movement **into** a place; **emigration** means movement **out** of a place.

Push and pull factors

- **Push factors** make the origin less attractive — unemployment, poverty, drought (सूखा), flood, conflict, or lack of services.
- **Pull factors** make the destination more attractive — jobs, education, better wages, urban amenities, and industrial hubs.
- Uttar Pradesh and Bihar are classic **out-migration** source states; Maharashtra, Delhi and Gujarat are classic **in-migration** destinations.

A **Census town** must satisfy three conditions together: population ≥ **5,000**, density ≥ **400** persons/km², and ≥ **75%** of male main workers in non-agricultural pursuits.

N.9 Population Composition (age, occupation, SC–ST)

Composition answers **who** the people are, not only how many.

Age composition

- Children (**0–14**), working age (**15–59**), and aged (**60+**) are the standard three bands.
- A high child share means a broad-base pyramid and higher future labour entry.
- **Demographic dividend** exists while the **15–59** share is high and dependency is manageable.
- **Dependency ratio** note compares young + aged dependents with the working-age population.

Occupational structure

- Workers are grouped as **primary** (agriculture and allied), **secondary** (industry), and **tertiary** (services).
- India still has a large primary workforce in Census note, but development shifts people toward secondary and tertiary work.
- Males are more visible in non-farm work; many women remain concentrated in primary occupations in rural areas.

Social composition (Census 2011 note)

- Scheduled Castes are about **16.6%** of India's population.
- Scheduled Tribes are about **8.6%** of India's population.
- Rural share is about **68.84%** and urban share about **31.16%**.

N.10 Urbanization

Urbanisation is the rising share of people living in **urban** places, plus the growth of towns and cities in number and size.

- India's Census **2011** urban share is **31.16%**; the rural share is about **68.84%**.
- A **statutory town** is notified under law as a municipality, corporation, cantonment or notified town area.
- A **census town** is not necessarily a municipality; it must pass the triple test of population ≥ **5,000**, density ≥ **400** persons/km², and ≥ **75%** of male main workers in non-agricultural pursuits.
- On the urbanisation curve used in standard notes, the **acceleration** stage is the **second stage**.
- Urbanisation and high literacy both help reduce the birth rate over time.
- Among religious groups, **Jains** have the highest urban share in Census 2011.
- Among States, **Goa** (गोवा) is classically taught as the most urbanised State by share; large northern States remain more rural.
- Uttar Pradesh's urban share in 2011 is about **22.3%**, well below India's **31.16%**, while its rural share is about **77.7%** against India's **68.84%**.
- Urban growth comes from natural increase inside towns, rural–urban migration, and reclassification of rural units as census or statutory towns.

Inline PYQ — UPPCS Prelims 2021, Q48

Acceleration stage in the urbanisation curve is the:

A. First stage

B. Second stage

C. Third stage

D. Fourth stage

Show answer

Ans: B — Acceleration = **second stage**.

Inline PYQ — UPPCS Prelims 2022, Q96

Which religious group in India has its highest urban population?

A. Hindu

B. Buddhist

C. Christian

D. Jain

Show answer

Ans: D — **Jains**.

N.11 Million Cities

A Census **million city / urban agglomeration** has population ≥ **10 lakh** (1 million).

- An **urban agglomeration (UA)** is a continuous urban spread that may include a city with its outgrowths and adjoining towns.
- Census **2011** counted **53** million-plus UAs, up from **35** in 2001.
- The largest UAs in the 2011 national list are led by **Greater Mumbai**, **Delhi** and **Kolkata**, followed by other metros such as Chennai, Bengaluru and Hyderabad.
- Hierarchy terms:
- A **Class I** town has population ≥ **1 lakh**.
- A **million city** has ≥ **10 lakh**.
- A **megacity** in UN usage is often ≥ **10 million**.
- A **conurbation** is a merged urban sprawl of several towns.
- Use the definition given in the stem.
- In Uttar Pradesh, **Kanpur** crossed 10 lakh in **1971** and **Lucknow** (लखनऊ) in **1981**.
- UP UA size order in 2011 is **Kanpur > Lucknow > Ghaziabad > Agra**.
- Million-city growth concentrates jobs, services and migrants, so these UAs pull strongly from surrounding rural districts.

N.12 Population Pyramid & Age Structure

A population pyramid is a double bar diagram of age–sex structure. Males are usually shown on the left and females on the right, with age bands stacked from young at the base to old at the top.

- Age **0–14** is the child / young band.
- Age **15–59** is the working-age band used for the **demographic dividend** fact in standard notes.
- Age **60+** is the aged band and is **not** that dividend definition.
- An **expanding** pyramid has a broad base and tapers upward — it signals high fertility and a youthful population.
- A **constrictive** or ageing pyramid has a narrower base and a relatively thicker upper part — it signals low fertility and longer survival into old age.
- A more rectangular / **stationary** shape sits between those two extremes when birth and death rates have both fallen.
- **Dependency ratio** compares young plus aged dependents with the working-age population.
- India remains youthful while the **15–59** share stays high; southern States show earlier ageing signals than the high-fertility northern belt.
- Uttar Pradesh's higher fertility history keeps a broader youthful base than many southern States in the same Census set.

Inline PYQ — UPPCS Prelims 2022, Q129

The demographic dividend is—

A. Working population of above 60 years

B. Working population of 14 to 50 years

C. 0–6 years population

D. Working population of 15 to 59 years

Show answer

Ans: D — Dividend = **15–59**.

N.13 Demographic Transition & Theories

Four named theories | Cannan · Henry George · Thompson · Malthus

Population theories explain **why numbers grow**, **when growth slows**, and **whether a given size is too large or too small** for the resources in use.

Theory	Propounder
Optimum Population	**Edwin Cannan**
Social mal-adjustment	**Henry George**

Theory	Propounder
Demographic Transition	**Thompson**
Population–food	**Malthus**

If a stem matches theory to person, use this table first. Other scholars appear in options as popularisers, critics, or parallel theories. They do **not** replace the four names above.

Malthus — Population–food (theory of population)

Thomas Malthus (1798, *Essay on the Principle of Population*) argued that population grows in **geometric** progression (1, 2, 4, 8…) while food grows in **arithmetic** progression (1, 2, 3, 4…).

- The Malthusian Theory is a theory of **population**. It is not a theory of poverty, unemployment, or the economy.
- When numbers outrun food, misery returns through **positive checks** that raise the death rate: famine, disease and war.
- **Preventive checks** lower the birth rate: delayed marriage, celibacy and moral restraint.
- Do not swap the two progressions. Geometric is population. Arithmetic is food.
- **Neo-Malthusians** later kept the food-pressure idea but stressed **birth control** as the main check.
- **David Ricardo**'s diminishing returns on land is often taught beside Malthus. The geometric/arithmetic pairing itself remains **Malthus**.

Optimum Population — Edwin Cannan

Edwin Cannan defined **optimum population** as the size at which, with given resources and technique, **income per head** is **maximum**.

- If actual population is **below** that size, the country is **underpopulated** and per-head income sits below the peak.
- If actual population is **above** that size, the country is **overpopulated** and per-head income falls.
- **Lionel Robbins**, **Hugh Dalton** and **A.M. Carr-Saunders** restated the same optimum idea. If the stem asks who **propounded** Optimum Population Theory, the name is still **Edwin Cannan**.
- Robbins spoke of the population that just makes **maximum returns** possible.
- Dalton spoke of the population that gives **maximum income per head**.
- Carr-Saunders spoke of the population that yields **maximum economic welfare**.
- Dalton called the gap from the optimum **maladjustment**, written as **M = (A − O) / O**, where A is actual population and O is optimum. If M is positive the country is overpopulated. If M is negative it is underpopulated. If M is zero it is at the optimum.
- Dalton's word **maladjustment** is **not** Henry George's Social Mal-adjustment Theory.

Social Mal-adjustment — Henry George

Henry George (*Progress and Poverty*) argued that poverty is a **social mal-adjustment**, not a natural Malthusian food-arithmetic law.

- He blamed **unequal land and wealth** (land monopoly and unjust distribution), not “too many people”.
- Do not pair this theory with Malthus, Cannan, Thompson or Dalton.

Demographic Transition — Thompson (Notestein)

Warren Thompson is the named propounder in the usual match list. **Frank Notestein** restated and popularised the same transition idea. **C.P. Blacker** is remembered for a **five-stage** version that adds a final **declining** stage.

The classic four-stage path is **high stationary → early expanding → late expanding → low stationary**.

Stage	Birth rate	Death rate	Growth
1 High stationary	High	High	Slow
2 Early expanding	High	Falls first	Rapid (explosion)
3 Late expanding	Falls	Low	Slows
4 Low stationary	Low	Low	Stable / very slow
5 Declining (Blacker)	Very low	Low, but can exceed births	Shrinks / ages

- Stage 1 stays high-high because of disease, poor health care and food shortage. Many notes illustrate it with pre-industrial / tribal societies.
- Stage 2 is the explosion stage because **death falls first** (health, sanitation, food) while births stay high. Urbanisation **acceleration** in the classic curve is linked to this **second** stage.
- Stage 3 is when **births fall** through education, urban living and family planning.
- Stage 4 is low births and low deaths in a mature industrial society.
- Stage 5 (when used) is ageing and possible natural **decrease**, as in several European countries and Japan.
- India is generally framed in the **later expanding** phase after the Stage 2–3 shift, not in Stage 1. It is not parked forever in the explosion stage.

Other scholars who appear as options

These names sit beside the four-theory table. They are parallel ideas or later restatements, not replacements for Cannan, Henry George, Thompson or Malthus.

Idea	Person
Surplus population under capitalism	**Karl Marx** (with Engels)
Population growth can intensify farming	**Ester Boserup**
Social capillarity (rise in status → fewer children)	**Arsène Dumont**
Five-stage transition including decline	**C.P. Blacker**
Fertility falls as density rises	**Sadler**

Idea	Person
Diet theory (better fed → lower fertility)	**Doubleday**
Energy to mind versus reproduction	**Herbert Spencer**
Optimum restated (not the first name)	Robbins / Dalton / Carr-Saunders
DTT restated (not the first name)	**Notestein**

- **Karl Marx** rejected Malthus's natural-law overpopulation. He treated a **surplus population** as a product of **capitalist** production and unequal distribution, not of geometric births versus arithmetic food.
- **Ester Boserup** argued the opposite pressure story. A larger population can **stimulate agricultural intensification** (more labour, shorter fallow, better technique) rather than only famine.
- **Arsène Dumont**'s **social capillarity** says people who want to rise in status tend to have **fewer children**. Do not confuse Dumont with Henry George.
- **Sadler** linked lower fertility with higher density.
- **Doubleday** linked lower fertility with a richer diet.
- **Herbert Spencer** argued that energy spent on mental development reduces energy left for reproduction.

Inline PYQ — UPPCS Prelims 2023, Q122

Match List-I with List-II and select the correct answer.

List-I (Theory)	List-II (Propounder)
A. Optimum Population Theory	1. Thompson
B. Social Mal-adjustment Theory	2. Malthus
C. Demographic Transition Theory	3. Edwin Cannan
D. Population–Food Supply Relationship Theory	4. Henry George

> Row order is **not** the answer — use the option codes.

A. A-(3), B-(4), C-(1), D-(2)

B. A-(2), B-(3), C-(4), D-(1)

C. A-(1), B-(2), C-(3), D-(4)

D. A-(4), B-(3), C-(1), D-(2)

Show answer

Logic: The stem tests four theory–person pairs. Cannan is optimum, not DTT. Henry George is social mal-adjustment, not Dalton's optimum formula. Thompson is DTT. Malthus is population–food. **Ans: A.** A-3 Cannan, B-4 Henry George, C-1 Thompson, D-2 Malthus.

Inline PYQ — UPPCS Prelims 2024, Q94

Malthus argued that the population grows in a **__ progression, while agricultural production/food supply grows in a** ____ progression.

A. Linear, Exponential

B. Exponential, Linear

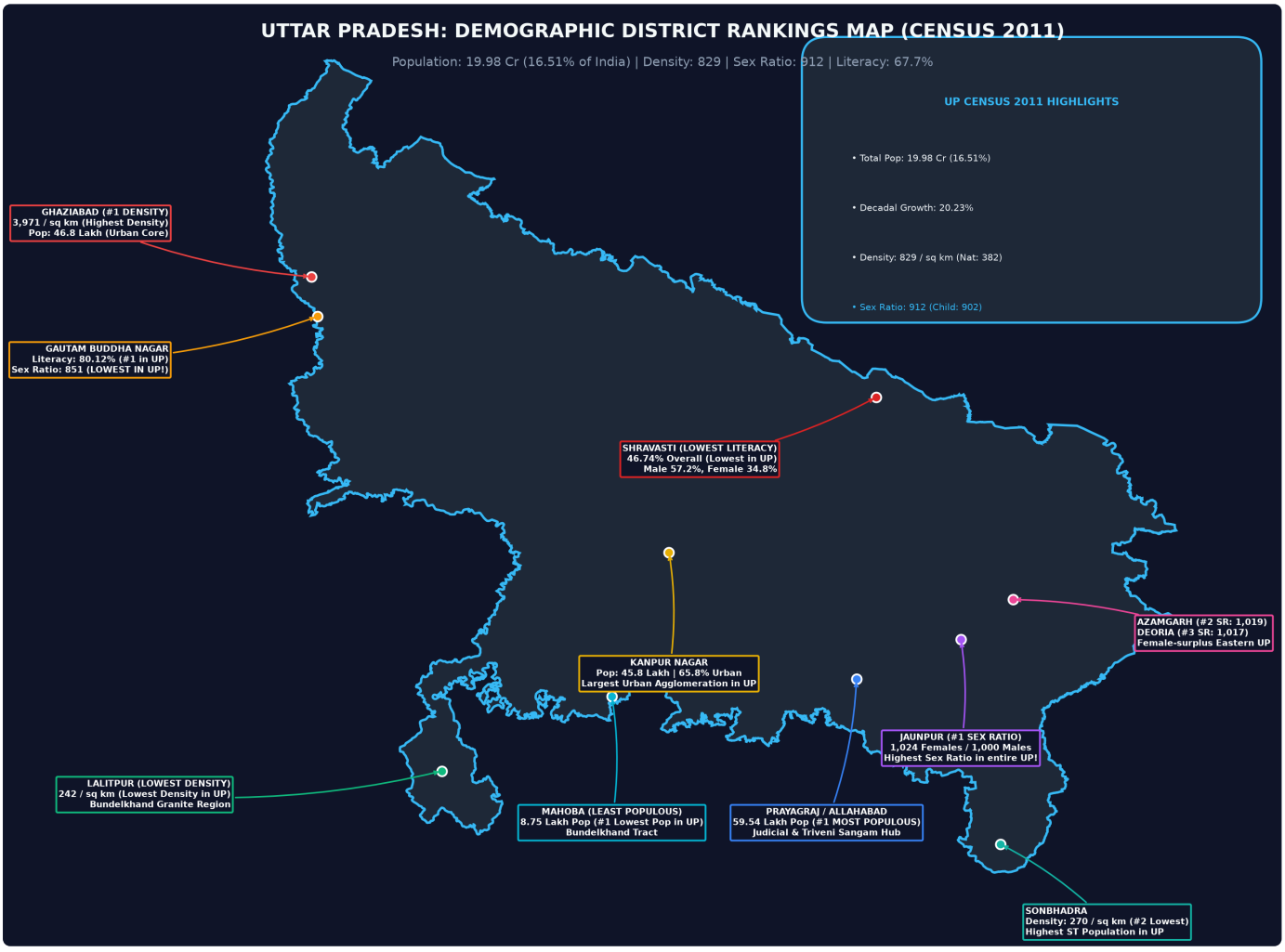
C. Arithmetic, Geometric

D. Geometric, Arithmetic

Show answer

Logic: The stem tests the exact Malthus wording. Option C swaps the two progressions. Linear/exponential wording is a paraphrase trap, not the standard pair. **Ans: D.** Population grows in **geometric** progression. Food grows in **arithmetic** progression.

N.14 Uttar Pradesh Focus



Uttar Pradesh is India's **most populous** state and also has the **largest rural** population in Census 2011.

- UP held about **16.5%** of India's population (~**19.98 crore** of India's **121.09 crore**).

- Density is **829** against India's **382**.
- Sex ratio is **912** against India's **943**.
- Literacy is about **67.7%** against India's **74.04%**.
- Urban share is about **22.3%** against India's **31.16%**. Rural share is about **77.7%** against India's **68.84%**.
- Decadal growth **2001–11** was about **20.23%**, above India's **17.64%**.
- **Prayagraj** is the most populous district in the 2011 note.
- **Ghaziabad** is the densest district.
- **Jaunpur** ranks among the better sex-ratio districts.
- **Shrawasti** has the lowest female literacy among UP districts.
- Million-city years: Kanpur **1971**, Lucknow **1981**.
- UA order 2011: **Kanpur > Lucknow > Ghaziabad > Agra**.
- Never treat TFR as a "per thousand" figure — that unit belongs to CBR.

Item	Uttar Pradesh 2011	India 2011
Rank	Most populous state; largest rural population	—
Population	**~19.98 crore** (**~16.5%** of India)	**121.09 crore**
Density	**829**	**382**
Sex ratio	**912**	**943**
Literacy	**~67.7%**	**74.04%**
Urban share	**~22.3%**	**31.16%**
Rural share	**~77.7%**	**68.84%**
Decadal growth 2001–11	**~20.23%**	**17.64%**
Female literacy floor (district)	**Shrawasti**	—
Million years	Kanpur **1971**, Lucknow **1981**	—
UA order	Kanpur > Lucknow > Ghaziabad > Agra	—